NEW TRIANGLES AT ±2: PANEL RESULTS, THE SHARPNESS OBSTRUCTION, AND THE CORRECTED COMPLEX

Sources: `notes/R7/` (`tri-cover`, `tri-orbifold-g`, `tri-pillowcase-g`, `tri-topology-g`, `prove`, `test`, `reducible-index`, `corrected/`). Target: all nontrivial K. $A_i = C(Y_2, P_i)$, $B_i = C(Y_{-2}, P_i)$, bd $= (\frac{1}{4}, 0)$.

1. Panel: four angles, no triangle meets the requirements

The requirements were (1) an injective connecting map, (2) of topological level exactly $-\frac{1}{16}$, supporting flat connections in the mixed case, and (3) exact at $T(2,-3)$ and 4_1.

angle	what is new	why it fails
cover	Δ_F: branch W' along the capped Seifert surfaces; middle term $C(S^3) = 0$ for all K.	$b^+ = 2g-1$, so the level is $(4g-3)/8 = +\frac{5}{8}$ at $g = 2$.
orbifold, pillowcase (found independently)	$F: (Y_2, P_\varepsilon) \to (Y_{-2}, P_{1-\varepsilon}) \to I(Y_0) \oplus I(Y_0, w)$. Its side map $\mathcal{E}_{\gamma_1} + \mathcal{E}_{\gamma_1'}$ has topological level *exactly* $-\frac{1}{16}$. It is the only rigid map of that level in the pillowcase.	Null for every K: each mixed point factors through an S^3 edge via a corner triangle of area $\frac{1}{16}$ ($\mathcal{E}_{\gamma_1} = \partial g^{ns}$). The third vertex is nonzero.
topology	$Y_{-2} = (Y_2)_{\lambda'}(-1)$, with λ' the Seifert push-off of K. This gives a triad $Y_2 \to Y_{-2} \to Y_0 \# \mathbb{RP}^3$ through one negative definite 2-handle, with form (-1). The singular maps along its core and cocore have $s = \frac{1}{2}$, so level exactly $-\frac{1}{16}$; their pillowcase line $H = \{b = \pm\frac{1}{4}\}$ passes through all four mixed points.	Same bundle only, and the third vertex is nonzero. The H-maps are not flat at bd. The triple point (L_2, H, L_{-2}) has zero area, which is the degenerate configuration of triangle-1 Prop. 3.1.

Pillowcase classification (proved in the Fukaya model). Every triangle with vertices L_2^ε and $L_{-2}^{1-\varepsilon}$ is the cone of a map spanned by the two mixed points.

- The backward cones have middle term $V_\alpha \oplus E_0 \oplus E_{1/2}$.
  - $\alpha = \frac{1}{4}$ gives the known triangles (I) and (II).
  - $\alpha \neq \frac{1}{4}$ gives Triangle(α). Its cap areas $4(\alpha - \frac{1}{8})^2$ and $4(\alpha - \frac{3}{8})^2$ differ, so the theory is non-monotone.
  - $\alpha \to 0$ gives the Farey route, with middle term 0 but level $\geq \frac{1}{8}$.
- The forward cone is F.

Nothing else exists.

2. The sharpness obstruction

Proposition (N2)**.** *Let $f\colon A_1 \to B_0$ be defined uniformly in K by counting, possibly broken, instantons on cobordisms or families that contain $E_K \times [0,1]$ compatibly with both ends. This covers surgery, singular, orbifold and lens-capped cobordisms, families, composites, Toda brackets and pillowcase polygons. If f is injective at $K = 4_1$, then $L_{\mathrm{top}}(f) > -\frac{1}{16}$.*

Proof.

(1) For 4_1 we have $\sigma = 0$, so the correction of §3 does not enter. Both complexes are perfect: A_1 has its bottom class at bd, level $\frac{1}{5}$, and $B_0 = \{z\}$ at level $\frac{11}{80} = \frac{1}{5} - \frac{1}{16}$.
(2) An injective map of level $-\frac{1}{16}$ must send x_{bd} to z through an instanton of energy 0, i.e. a flat connection.
(3) A flat connection restricts to a single representation of $\pi_1(E_K \times I)$. But bd and z are different representations: bd is binary dihedral with $a = \frac{1}{4}$, and z is not. □

Consequences.

- In this toolkit, no single injective map proves the sharp bound $m_2 \geq \frac{1}{16}$.
- If levels lie in $\frac{1}{16}\mathbb{Z}$, as every row of the arithmetic table does, the best possible outcome is $m_2 > 0$, coming from a level-0 map plus positivity of energy.
- A flat connection realising equality must change the E_K-representation. It therefore lives on a cobordism that does *not* contain $E_K \times I$ compatibly.
- It also cannot pass through a vertex carrying only reducibles, if that vertex is reached by 2- and 3-handles (topology, O3). The holonomy image of bd is dihedral (D_3 for $T(2,-3)$, D_5 for 4_1) and has no element of order 4, so a γ_1 end never occurs.

In the data the equality is carried by $b' \circ a'$: a' is flat at bd, and b' has energy exactly $\frac{1}{16}$, the area of the mixed triangle.

Escapes left.

- A cobordism $Y_2 \to Y_{-2}$ not containing $E_K \times I$. Its flat part must change representation, e.g. by adding 1-handles.
- A non-topological cancellation.
- Settling for a strict bound $m_2 > 0$. This needs a non-null mixed map of topological level 0 that is nonzero on the bottom class. The known ones, W'_{odd} and $b \circ a$, are null.

3. **The corrected complex for** $(Y_{\pm 2}, w)$

Reducible index (`reducible-index.md`).

- On $Y_1 \to (Y_2, w)$, the twisted reducible $A_k = \mathbb{R} \oplus L$ with $c_1(L) = (2k+1)e$ has $e^2 = -\frac{1}{2}$ and energy $(2k+1)^2/8$.
- Its L-part has complex index $\frac{(2k+1)^2-1}{2} + \frac{\rho_\chi(Y_2)}{2}$, where $\rho_\chi(Y_{\pm 2}) = \sigma(K)$ in the APS convention (Casson–Gordon).
- So the minimal reducible ($\kappa = \frac{1}{8}$) is obstructed with cokernel $\mathbb{C}^{|\sigma|/2}$ on one of the two edges at $(Y_{\pm 2}, w)$.
- This gives $|\sigma|/2$ suspensions, and none when $\sigma = 0$, in particular when $\Delta_K = 1$.

Your rule, applied. I added $|\sigma|/2$ generators with Chern–Simons equal to that of ζ. I read the degrees as $\text{ind}(\zeta \to \cdot)$. Read as $\text{ind}(\cdot \to \zeta)$ instead, the excess cases are inconsistent: the new and old generators have the same parity, so the rank cannot come out right. In the deficit cases the new generators survive. In the excess cases each one cancels an old generator, constrained by the $\mathbb{Z}/8$ grading and by $E > 0$. Data: the 105 torus knots with $pq \leq 120$, in both chiralities.

drop	mirror knots ($\sigma > 0$)	positive knots ($\sigma < 0$)
$\lambda_1(Y_2) - \lambda_0(Y_{-2})$	$\geq \frac{1}{16}$, equality only at $T(2,-3)$	< 0 for all 105, minimum -0.234 at $T(3,40)$
$\lambda_0(Y_2) - \lambda_1(Y_{-2})$	$\geq \frac{1}{9}$ ($T(4,-5)$)	depends on the cancellations; best case ≥ 0.131
$\lambda_0(Y_2) - \lambda_0(Y_{-2})$	$\geq \frac{3}{32}$, unchanged	≥ -0.109, unchanged (DLME)
$\lambda_1(Y_2) - \lambda_1(Y_{-2})$	≥ 0.1225	depends on the cancellations; best case ≥ 0.006
$\min_i \lambda_i(Y_2) - \min_i \lambda_i(Y_{-2})$	≥ 0.1244	depends on the cancellations; best case ≥ 0.006

For positive knots the surviving new generators of (Y_2, w) sit at levels $\frac{1+|\sigma|}{8}, \frac{|\sigma|-1}{8}, \ldots,$ ending at *exactly* $\frac{3}{8}$. So $\lambda_1(Y_2) = \frac{3}{8} < \lambda_0(Y_{-2})$.

Caveat. This depends on two normalisations. An offset of 2 in degree moves the bottom generator to $\frac{5}{8}$. Taking Chern–Simons to be 0 in the absolute normalisation, rather than at ζ, moves it to $\frac{1}{4}$. The mirror-knot column depends on the same choices; in this reading no mirror-knot drop decreases.

4. **Calibration for general knots**

- The sharp, non-strict $\frac{1}{16}$ cannot come from any map built on $E_K \times I$ (N2). Equality at 4_1 is witnessed by positive energy on a triangle face.
- What such maps can give is a strict bound from a level-0 map. Even that needs a non-null mixed map that is injective on the bottom class, and none is known.
- The $-\frac{1}{16}$ maps that do exist are F's side map, $\mathcal{E}_{\gamma_1}$ and the $s = \frac{1}{2}$ singular maps of λ'. All are null, go to or from a singular theory, or are not flat at bd.